

Kang Zheng

Ph.D. Candidate in Geospatial Artificial Intelligence

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Profile

Kang Zheng is a Ph.D. candidate working at the intersection of artificial intelligence and geoscientific research. His research spans land-use and land-cover mapping, road element extraction, PM2.5 concentration assessment, and emerging GeoAI applications based on foundation models and multimodal fusion. He has published 19 papers, including 14 as first or corresponding author and 11 in CAS Zone 2 or above journals.

Education

- 2023-2027, Wuhan University, Ph.D. in Resources and Environment (Surveying and Mapping Engineering), advised by Prof. Fu Ren.
- 2026-2027, Politecnico di Milano, CSC visiting Ph.D. research in Geographic Information Science, advised by Prof. Maria Antonia Brovelli.
- 2020-2023, Henan University, M.Sc. in Cartography and Geographic Information System, advised by Prof. Haiying Wang.
- 2016-2020, Hunan City University, B.Sc. in Geographic Information Science.

Publications

- 2026. Multi-scale feature extraction of road marking from front-view images through hybrid convolutional and vision transformer for high-definition map. K. Zheng, J. Wang, Y. Chen, J. Zhan, and Z. Liu. Engineering Applications of Artificial Intelligence, 179, 115165. doi: [10.1016/j.engappai.2026.115165](#) (First author; JCR Q1; CAS Zone 1 Top; IF 9.0)
- 2025. Daily high-resolution PM2.5 mapping using spatiotemporal CNN-transformer-KAN model. Z. Liu, K. Zheng*, Y. Song, and J. Zhang. International Journal of Applied Earth Observation and Geoinformation, 144, 104900. doi: [10.1016/j.jag.2025.104900](#) (First corresponding author; JCR Q1; CAS Zone 1 Top; IF 8.6)
- 2026. Spatiotemporal Mapping of PM2.5 Exposure in China (2009-2020) Using PMMamba Model: Impacts on Vulnerable Populations and Regional Inequality. Z. Liu, K. Zheng*, J. Wang, J. Zhang, and Y. Song. Science of Remote Sensing, 100436. doi: [10.1016/j.srs.2026.100436](#) (First corresponding author; JCR Q1; CAS Zone 2; IF 7.1)
- 2026. Efficient road marking extraction via cooperative enhancement of foundation models and Mamba. K. Zheng and F. Ren. Displays, 92, 103351. doi: [10.1016/j.displa.2026.103351](#) (First author; JCR Q1; CAS Zone 2; IF 3.4)
- 2026. Frequency and Prompt Learning Cooperation Enhanced Mamba for Remote Sensing Semantic Segmentation. K. Zheng et al.. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 19, 8463-8477. doi: [10.1109/JSTARS.2025.3607777](#) (First author; JCR Q1; CAS Zone 2; CAS Emerging Zone 2 Top; IF 6.3)
- 2025. Enhancing Remote Sensing Semantic Segmentation Accuracy and Efficiency Through Transformer and Knowledge Distillation. K. Zheng et al.. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 18, 4074-4092. doi: [10.1109/JSTARS.2025.3525634](#) (First author; JCR Q1; CAS Zone 2 Top; IF 4.7)
- 2024. Estimating the spatiotemporal distribution of PM2.5 concentrations in Tianjin during the Chinese Spring Festival: Impact of fireworks ban. Z. Liu, K. Zheng*, S. Bao et al.. Environmental Pollution, 361, 124899. doi: [10.1016/j.envpol.2024.124899](#) (First corresponding author; JCR Q1; CAS Zone 2 Top; IF 7.6)
- 2024. Forecasting daily PM2.5 concentrations in Wuhan with a spatial-autocorrelation-based long short-term memory model. Z. Liu, C. Ge, K. Zheng* et al.. Atmospheric Environment, 331, 120605. doi: [10.1016/j.atmosenv.2024.120605](#) (First corresponding author; JCR Q1; CAS Zone 2; IF 4.4)
- 2024. CREVNet: a Transformer and CNN-based network for accurate segmentation of ice shelf crevasses. K. Zheng, Q. Li, Z. Wang et al.. IEEE Geoscience and Remote Sensing Letters, 1-1. doi: [10.1109/LGRS.2024.3407860](#) (First author; JCR Q1; CAS Zone 3; IF 4.0)
- 2023. An improved land use classification method based on DeepLab V3+ under GauGAN data enhancement. K. Zheng, H. Wang*, F. Qin, C. Miao, and Z. Han. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 16, 5526-5537. doi: [10.1109/JSTARS.2023.3278862](#) (First author; JCR Q1; CAS Zone 2 Top; IF 4.715)
- 2022. A Land Use Classification Model Based on Conditional Random Fields and Attention Mechanism Convolutional Networks. K. Zheng, H. Wang*, F. Qin, and Z. Han. Remote Sensing, 14(11), 2688. doi: [10.3390/rs14112688](#) (First author; JCR Q1; CAS Zone 2 Top; IF 5.349)
- 2022. Using Multiple Sources of Data and "Voting Mechanisms" for Urban Land-Use Mapping. K. Zheng, H. Zhang, H. Wang*, F. Qin, Z. Wang, and J. Zhao. Land, 11(12), 2209. doi: [10.3390/land11122209](#) (First author; JCR Q1; CAS Zone 2; IF 3.905)
- 2026. 一种融合视觉状态空间模型与小波变换的车载图像轻量级道路要素提取网络. 郑康, 任福. 武汉大学学报 (信息科学版), Advance online publication. doi: [10.13203/j.whugis20250175](#) (First author; EI)
- 2026. 融合基础模型与多颜色空间的道路低照度图像增强及要素提取. 郑康, 陈亮, 任福, 杜清运, 田文涛, 廖梦帆, 张莹. 地球信息科学学报, 28(7), 2246-2261. doi: [10.12082/dqxxkx.2026.260107](#) (First author; EI)

- 2022. Deriving Urban Boundaries of Henan Province, China, Based on Sentinel-2 and Deep Learning Methods. X. Li, K. Zheng, F. Qin, H. Wang, and C. Zhao. Remote Sensing, 14(15), 3752. doi: [10.3390/rs14153752](https://doi.org/10.3390/rs14153752)
- 2024. Assessing landslide susceptibility based on the random forest model and multi-source heterogeneous data. M. Li, H. Wang, J. Chen, and K. Zheng. Ecological Indicators, 158, 111600. doi: [10.1016/j.ecolind.2024.111600](https://doi.org/10.1016/j.ecolind.2024.111600)
- 2024. 自动驾驶高精度地图特征分析及发展展望. 杜清运, 况路路, 任福, 刘江涛, 冯昶, 陈卓宁, 张淳聪, 郑康, 李智程. 地球信息科学学报, 26(1), 15-24. doi: [10.12082/dqxxkx.2024.240054](https://doi.org/10.12082/dqxxkx.2024.240054)
- 2025. 多源数据融合的内河航道水陆地形建模方法. 盛兆阳, 李晓雨, 郑康, 柯鉴, 任福. 时空信息学报, 32(5), 527-537. doi: [10.20117/j.jsti.202505003](https://doi.org/10.20117/j.jsti.202505003)
- 2026. 基于开放词汇语义分割的大范围城市绿地无监督提取. 丁婕玉, 郑康, 杜清运, 任福. 地球信息科学学报, 28(7), 1958-1973. doi: [10.12082/dqxxkx.2026.260095](https://doi.org/10.12082/dqxxkx.2026.260095)

Research Projects

- Completed. GIS-related Project. Completed project with a total funding of RMB 886,000; participant.
- 2024-Present. Key Technologies for a Spatiotemporal Data Infrastructure Platform Integrating Crowdsourced Data. National Key Research and Development Program of China, No. 2024YFB2505800; ongoing project with a total funding of RMB 120 million; participant.
- 2020-Present. Intelligent Spatiotemporal Big Data Analysis, Mining, and Visualization Platform for High-resolution Earth Observation Applications. Major project of the High-resolution Earth Observation System under the State Administration of Science, Technology and Industry for National Defense, No. 80-Y50G19-9001-22/23; ongoing project with a total funding of RMB 5.2 million; participant.
- 2023-2024. Application-oriented Research on Base Maps for Intelligent Connected Vehicles. Project funded by the Shenzhen Municipal Planning and Natural Resources Bureau, No. SZDL2023001359; completed project with a total funding of RMB 1.24 million; participant.
- 2020-2023. Fine-scale Simulation of Urban Growth Boundaries in the Central Plains Urban Agglomeration Using Machine Learning and Swarm Intelligence Algorithms. Project funded by the Natural Science Foundation of Henan Province, No. 202300410096; completed project with a total funding of RMB 100,000; participant.
- 2020. Urban Physical Examination Project of Zhengzhou. Participated in the 2020 urban physical examination project of Zhengzhou; completed.
- 2021. Urban Physical Examination Project of Zhengzhou. Participated in the 2021 urban physical examination project of Zhengzhou; completed.
- 2022. Urban Physical Examination Project of Anyang. Participated in the 2022 urban physical examination project of Anyang; completed.

Academic Service and Other Achievements

- Reviewer for more than ten journals, including GIS & Remote Sensing, IEEE Transactions on Geoscience and Remote Sensing, Engineering Applications of Artificial Intelligence, and Expert Systems With Applications; IEEE Student Member.
- Chinese invention patent: A land-use classification method and system based on data augmentation, ZL 2022 1 1678876.2.
- Additional experience: President of the college student union from July 2018 to July 2019; ranked sixth in a Henan provincial scientific and technological achievement in 2021.